ExpNo – 9 Experiment to understand matplotlib library use cases in data science using visualization

Program:

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

%matplotlib inline

data=pd.read_csv('Iris_Dataset.csv')

data.info()

data.describe()

data.value_counts('variety')

sns.countplot(x='variety',data=data,)

plt.show()

dummies=pd.get_dummies(data.variety)

FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)

FinalDataset.head()

sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)

sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)

sns.pairplot(data,hue='variety',height=3);

plt.show()

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();

plt.show();
```

```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();

plt.show();
```

Output:

```
<class 'pandas.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

 #  Column        Non-Null Count  Dtype

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 0  Id            150 non-null    int64

 1  SepalLengthCm  150 non-null    float64

 2  SepalWidthCm   150 non-null    float64

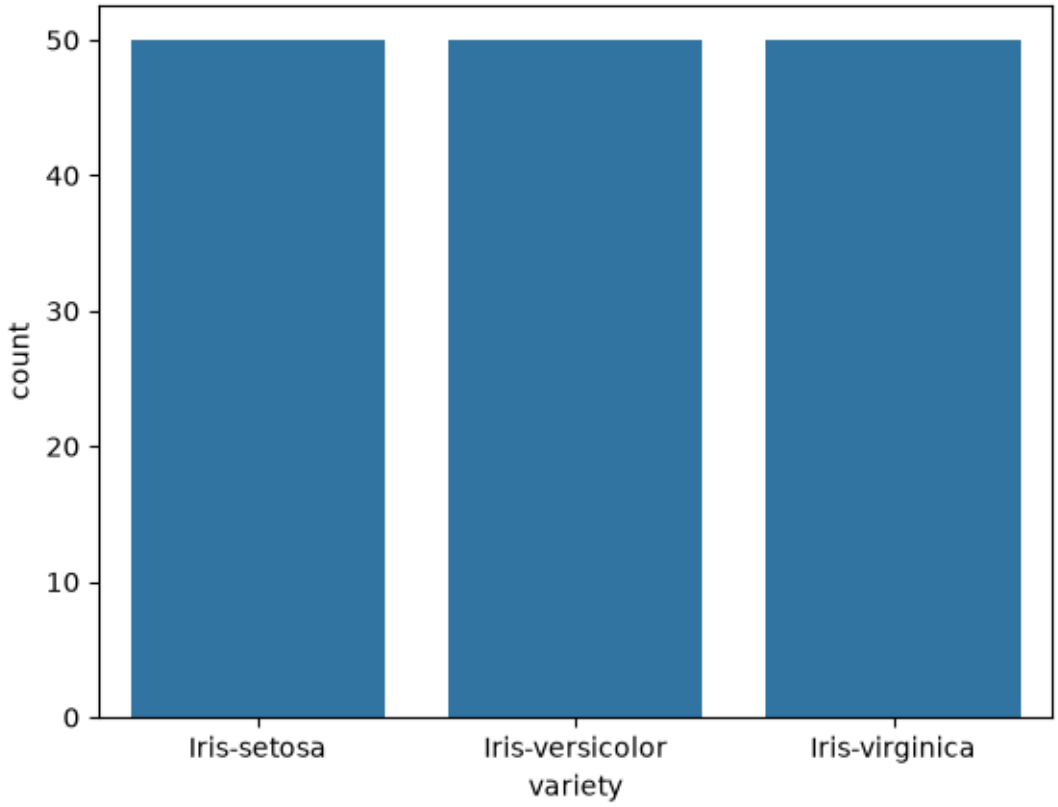
 3  PetalLengthCm  150 non-null    float64

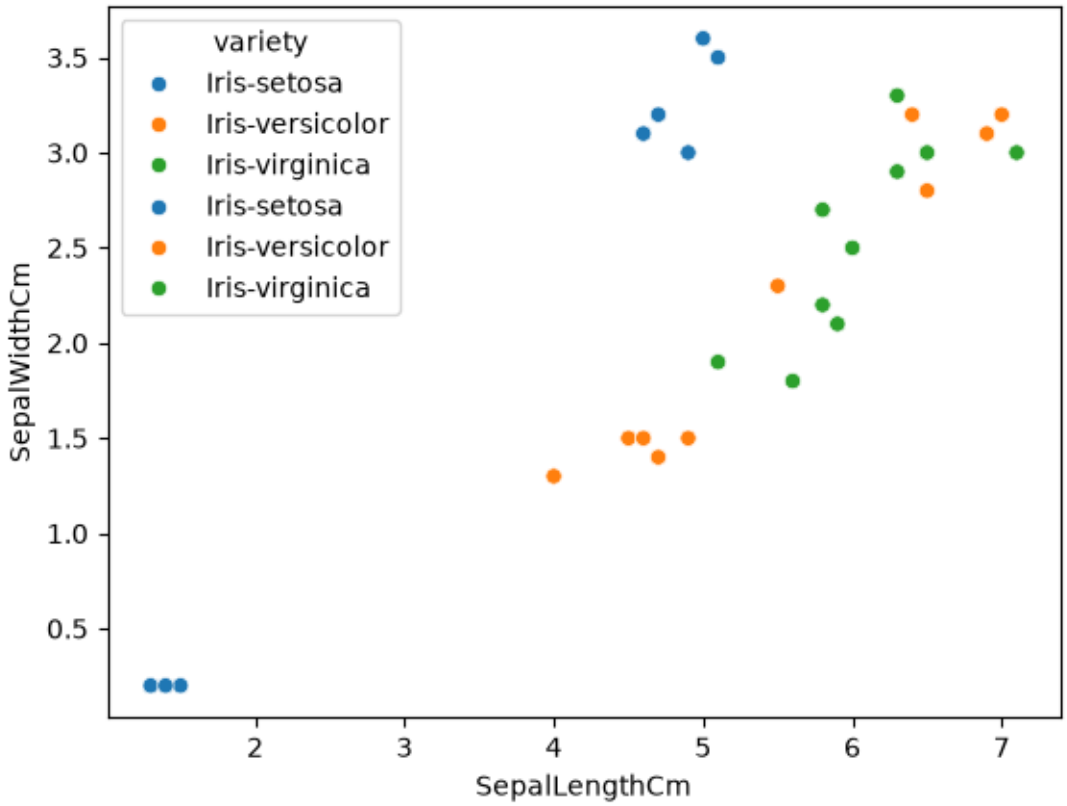
 4  PetalWidthCm   150 non-null    float64

 5  variety       150 non-null    str

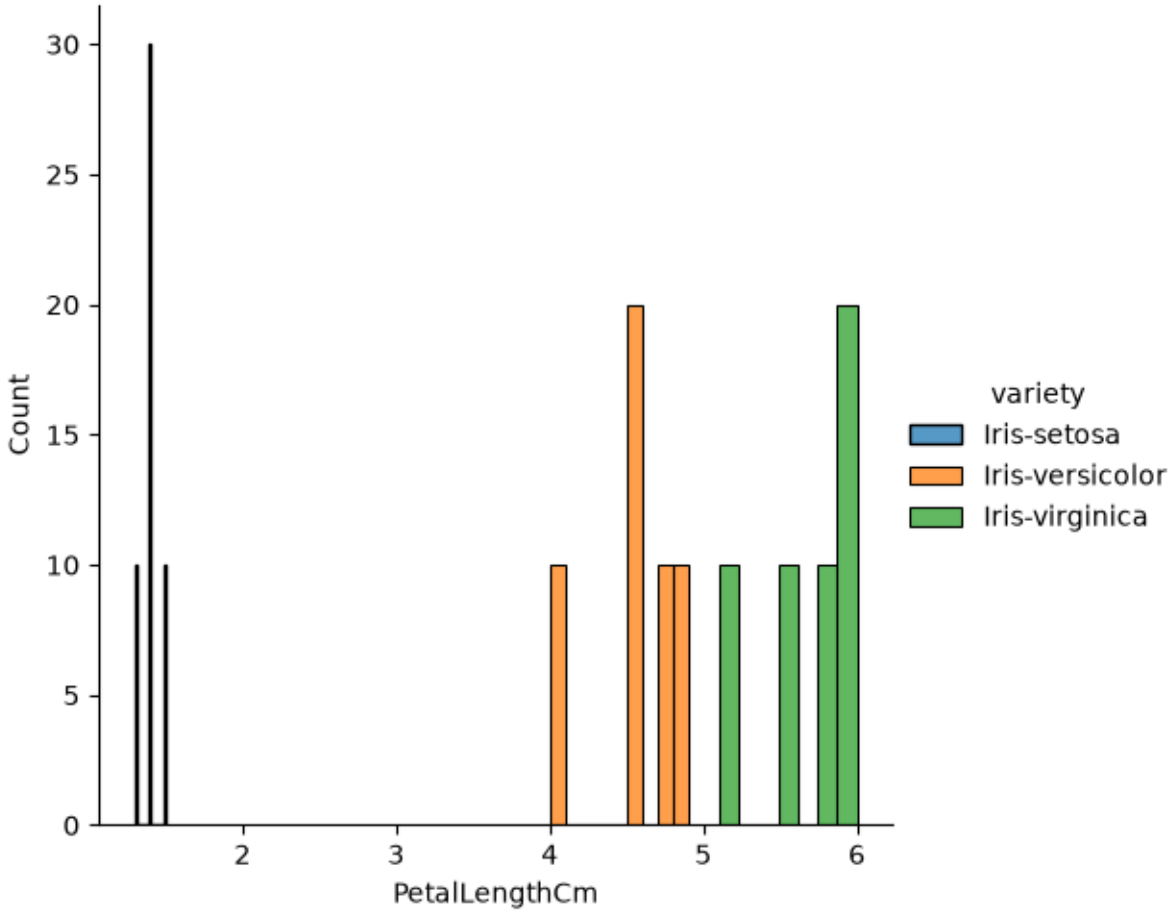
dtypes: float64(4), int64(1), str(1)

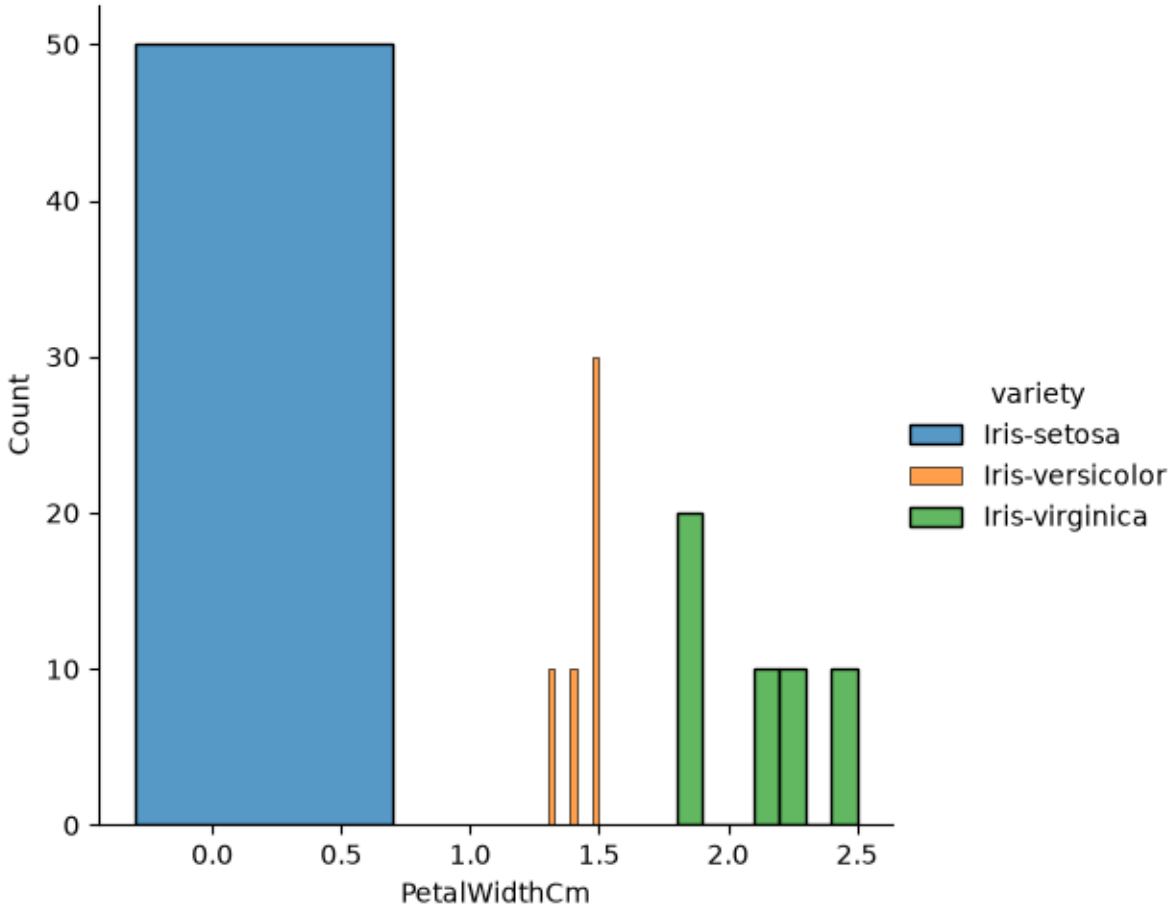
memory usage: 7.2 KB
```

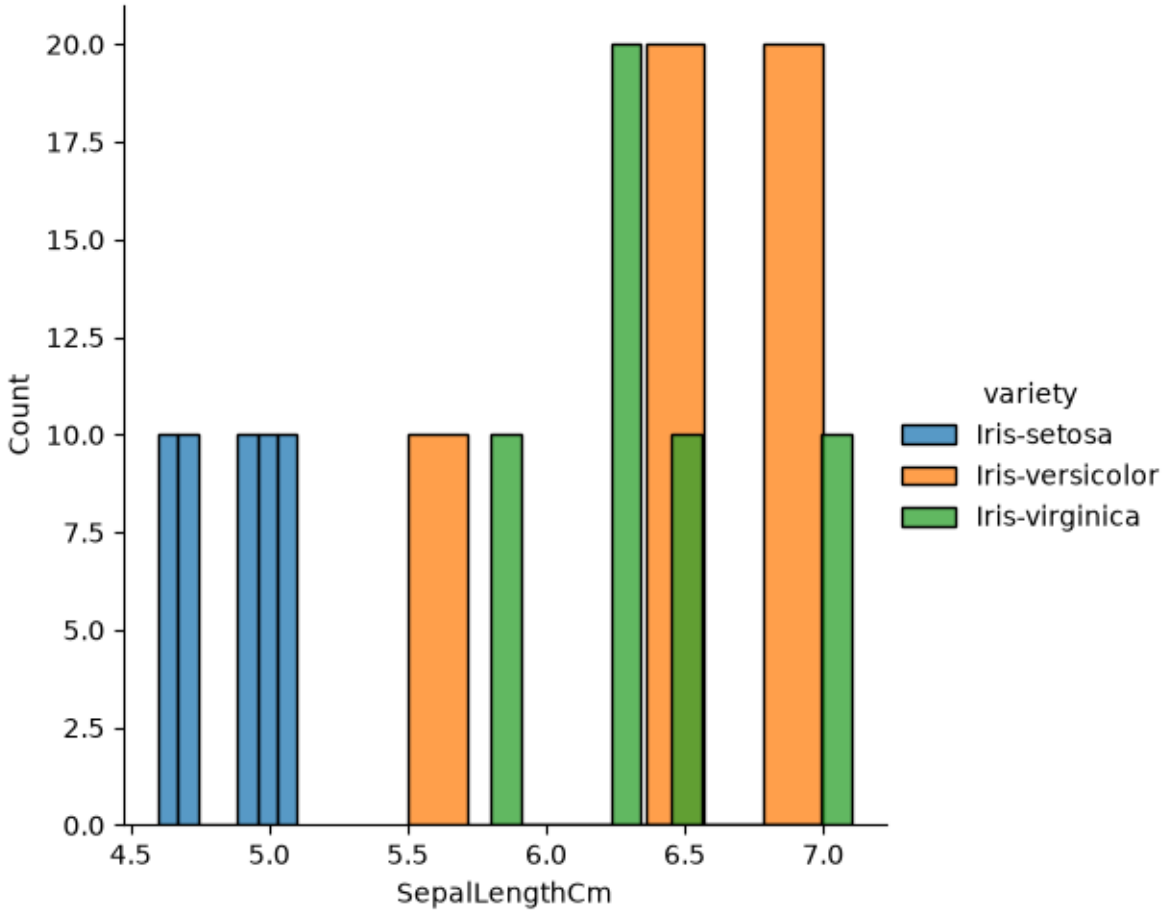

50
40
30
20
10
0
count
Iris-setosa
Iris-versicolor
Iris-virginica
variety

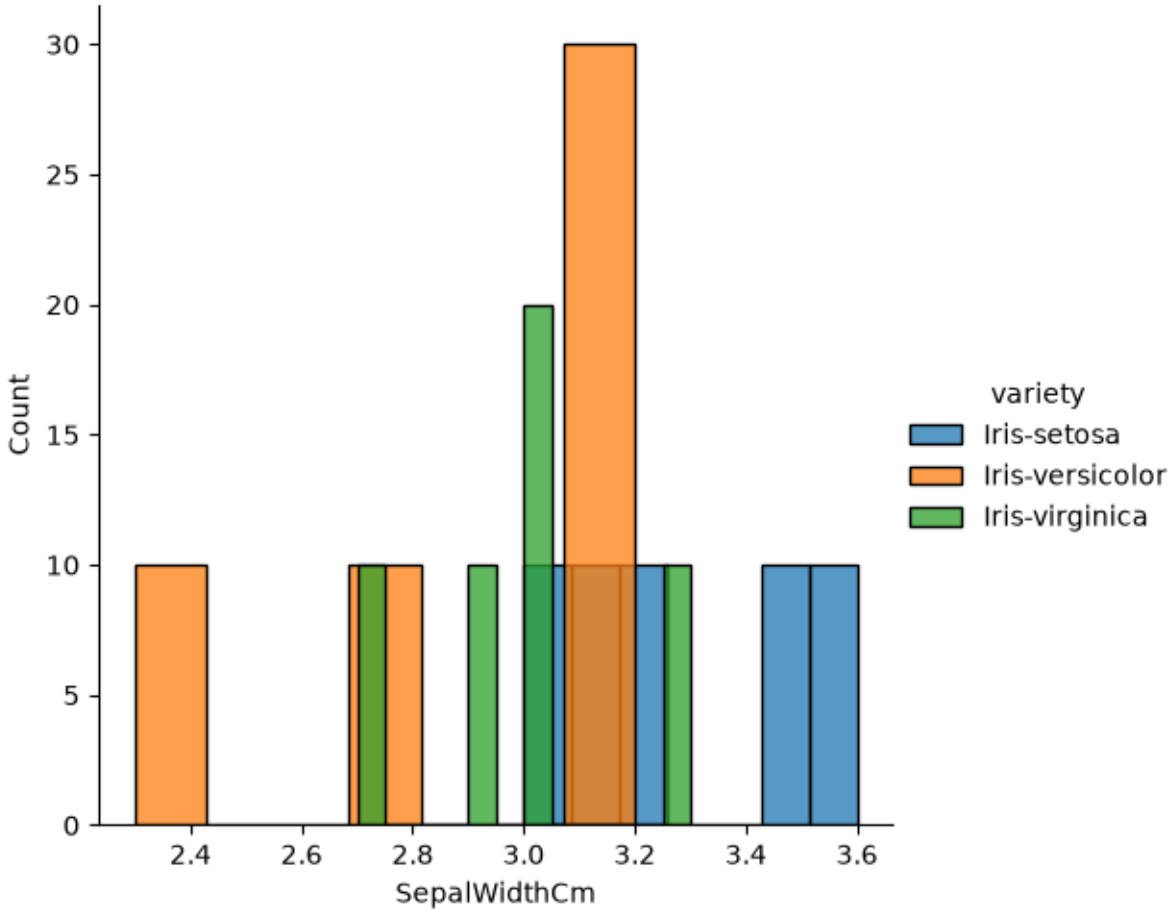

variety
Iris-setosa
Iris-versicolor
Iris-virginica
Iris-setosa
Iris-versicolor
Iris-virginica
3.5
3.0
2.5
2.0
1.5
1.0
0.5
SepalWidthCm
2
3
4
5
6
7
SepalLengthCm


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


30
25
20
15
10
5
0
Count
2
3
4
5
6
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


50
40
30
20
10
0
Count
0.0
0.5
1.0
1.5
2.0
2.5
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Count
4.5
5.0
5.5
6.0
6.5
7.0
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


30
25
20
15
10
5
0
Count
2.4
2.6
2.8
3.0
3.2
3.4
3.6
SepalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica